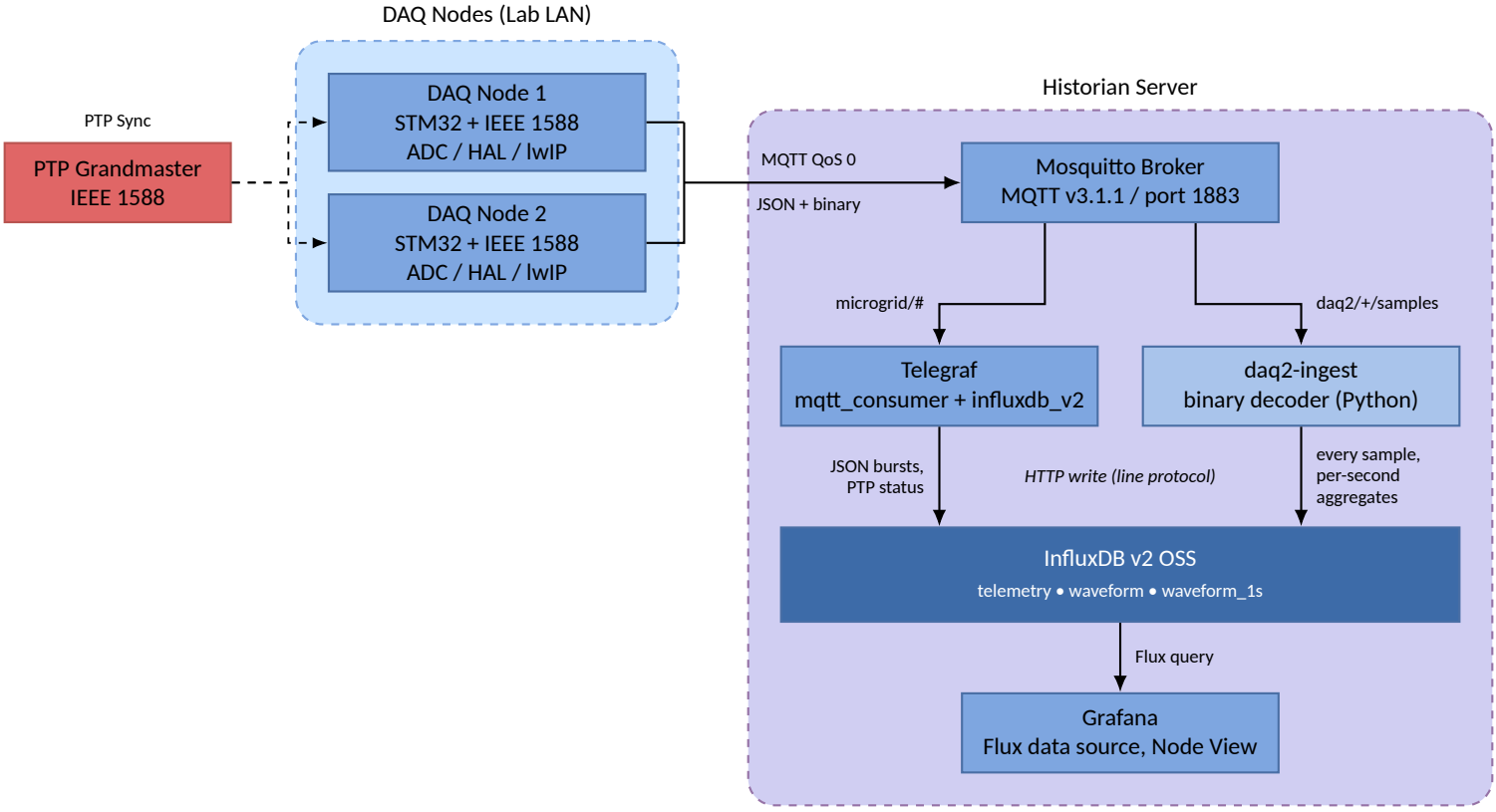
DAQ Nodes (Lab LAN)
PTP Sync
PTP Grandmaster
IEEE 1588
DAQ Node 1
STM32 + IEEE 1588
ADC / HAL / lwIP
DAQ Node 2
STM32 + IEEE 1588
ADC / HAL / lwIP
Historian Server
MQTT QoS 0
JSON + binary
Mosquitto Broker
MQTT v3.1.1 / port 1883
microgrid/#
daq2/+/samples
Telegraf
mqtt_consumer + influxdb_v2
daq2-ingest
binary decoder (Python)
JSON bursts,
PTP status
HTTP write (line protocol)
every sample,
per-second
aggregates
InfluxDB v2 OSS
telemetry • waveform • waveform_1s
Flux query
Grafana
Flux data source, Node View